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DETAIL SPECIFICATION

BURNERS, STEAM AND OIL PRESSURE, ATOMIZING

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the straight mechanical and steam and oil pressure atomizing burners for use in fossil-fueled propulsion boilers aboard naval ships. Burner assemblies consist of the atomizer, air register, and safety shutoff devices.

1.2 Classification. Burners are of the following types and classes as specified (see 6.2).

1.2.1 Types.

a. Type A – Straight pressure atomizing fuel oil burners (only for use for cold plant light-off [air not available] or if there has been a casualty to the atomizing steam system).

b. Type C – Steam and oil pressure atomizing burners.

1.2.2 Classes.

a. Class 1 – For mounting through boiler furnace double casing.

b. Class 2 – For mounting through boiler furnace double casing with air register hinged to provide for furnace access.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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FEDERAL SPECIFICATIONS

FF-N-836 - Nut: Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding and Single Ball Seat

FF-S-85 - Screw, Cap, Slotted and Hexagon Head

FF-S-86 - Screw, Cap, Socket-Head

FF-S-92 - Screw, Machine, Slotted, Cross-Recessed or Hexagon Head

FF-S-200 - Setscrews: Hexagon Socket and Spline Socket, Headless

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-1222 - Studs, Bolts, Screws and Nuts for Applications Where a High Degree of Reliability is Required; General Specification for

MIL-DTL-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5

MIL-DTL-15024 - Plates and Tags for Identification of Equipment, General Specification for

MIL-P-15024/5 - Plates, Identification

MIL-DTL-16884 - Fuel, Naval Distillate

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) DRAWINGS

221-7022537 - Sprayer Plate – Improved Wallsend Type Steam Atomizer (LHA/LHD/FF – 1078)

803-5959293 - Typical Arrangement of Fuel Oil Burner Leads for Boiler

Copies of these documents are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. Copies of these documents may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an account for drawing access, send an email to nnsy_jedmics_nsedr_helpdesk@us.navy.mil.

NAVAL SURFACE WARFARE CENTER PHILADELPHIA DIVISION (NSWCPD) DRAWINGS

4168 - Sprayer Plates – Racer-Navjet Stream Atomizer

Copies of this document are available from the applicable repositories listed in S0005-AE-PRO-010/EDM, which can be obtained online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. Copies of this document may also be obtained from the Naval Ships Engineering Drawing Repository (NSEDR) online at https://199.208.213.105/webjedmics/index.jsp. To request an account for drawing access, send an email to nnsy_jedmics_nsedr_helpdesk@us.navy.mil.

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at https://www.gsaglobalsupply.gsa.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 - Pipe Flanges and Flanged Fittings: NPS ½ through NPS 24, Metric/Inch Standard

(Copies of this document are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A105/A105M - Standard Specification for Carbon Steel Forgings for Piping Applications

ASTM A600 - Standard Specification for Tool Steel High Speed

ASTM E18 - Standard Test Methods for Rockwell Hardness of Metallic Materials

(Copies of these documents are available online at www.astm.org.)

SAE INTERNATIONAL

SAE-AMS-H-6875 - Heat Treatment of Steel Raw Materials

SAE-AMS-QQ-S-763 - Steel, Corrosion Resistant, Bars, Wire, Shapes, and Forgings

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 General construction. Construction shall be light and compact to the greatest extent practicable and consistent with requirements herein. Fabrication welding and inspection shall be in accordance with S9074-AR-GIB-010/278.

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3.2.1 General description. Each burner specified herein shall consist of an adjustable air register (to include diffuser), atomizer assembly, and safety shutoff device. Burners shall be constructed to prevent buckling, warping, or jamming of parts that require adjustments. Material shall be of a Technical Authority (see 6.6) approved type for the conditions encountered. The material of burner parts subjected to radiant heat (up to 1600 °F) from the furnace shall be of Technical Authority approved heat-resisting alloy. Burners shall burn Navy distillate fuel in accordance with MIL-DTL-16884 or JP-5 aviation turbine fuel in accordance with MIL-DTL-5624. Burner registers shall be provided with at least one observation port, having a lens of cobalt blue heat-resisting glass. At least one port on the register shall be usable as an insertion port for an electric ignitor or for the lighting-off torch for emergency light-offs. Provisions shall be incorporated to permit the atomizer to be removed from the air register and replaced without the use of a tool and to prevent discharge of furnace gases into the fireroom when the atomizer assembly is removed from the burner. The fuel oil and steam (for type C burners) can be expected to be supplied to the burner by means of branch line connections from the fuel oil supply and steam manifolds to the burner.

3.2.2 Shutoff devices. The following means for shutting off the fuel oil and steam shall be provided for each burner in addition to the flanged root valves provided between the burner and supply manifolds:

a. A safety shutoff device including separate steam and oil valves that shall be placed in the closed position before the atomizer assembly can be removed from the burner. The shutoff device shall have zero leakage when pressurized with Navy fuel at maximum system oil pressure with the burner installed or removed with shutoff valves closed.

b. Spring-loaded check valves that open only when the atomizer assembly is installed properly and closes when the atomizer assembly is loosened or removed from the burner.

c. For type C burners, the safety shutoff device shall incorporate the capability of purging steam through the atomizer with the oil valve shut. The steam and oil valves shall be interlocked so that it is not possible to open the oil valve with the steam valve closed.

3.2.3 Burner lead connections. Burner lead connections shall be forged carbon steel raised-face socket-welded flanges in accordance with ASTM A105/A105M and ASME B16.5 arranged in accordance with 803-5959293. Flange holes shall straddle the centerline.

3.2.4 Atomizer and diffuser assembly setting and adjustments. Provisions shall be made for adjusting and locking the atomizer assembly in proper position with respect to the diffuser to maintain the optimum atomizer tip setting. The burner shall also include an adjustment for positioning the atomizer and diffuser assembly relative to the throat ring or inner burner front wall (as applicable). The adjustment shall be lockable at either the inner or outer stop setting.

3.2.5 Allowance for movement and withdrawal. Oil burners shall be engineered for relative movement between the connections to the inner and outer burner front casing plates, in such a way that the atomizers and air registers cannot be thrown out of alignment by warping or expansion of the casings under high temperature conditions. Provisions shall be made to prevent accumulations of fuel oil drip between inner and outer casings under all conditions of operation. The construction of the burners shall permit convenient removal of the entire burner assembly through the burner front plate.

3.2.6 Air register. Air register dimensions shall be as specified (see 6.2). The air doors in the air register shall be capable of moving under all operating conditions. The air doors shall cycle from closed position to fully open position with less than one turn of the air door operating lever or handle.

3.2.7 Hinged air registers. The hinged air registers specified for class 2 burners shall be so constructed that the burner can be swung out to provide access to the furnace through the burner opening and be swung back into position.

3.2.8 Capacity to weight factor. Each burner at rated capacity shall deliver at least 8 pounds of fuel oil per hour per pound of burner weight. For calculations, all parts of the burner necessary for satisfactory operation are considered part of the burner weight except the burner throat refractory material, the retaining devices for this refractory material, and the branch fuel oil and steam lines.

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3.2.9 Capacity to area factor. Each burner at rated capacity shall deliver not less than 3.4 pounds of oil per hour per square inch of circular area covered by the closed burner (air doors shut) at the burner inner front plate (exclusive of holding ears or points in the case of polygonal base rings). This ratio shall apply to burners of capacity as great as 1,000 pounds per hour but shall be increased 0.1 pound of oil per hour per square inch for each 100-pound per hour increase in burner capacity.

3.2.10 Low capacity operation. Type A burners shall maintain a stable flame pattern down to a supply fuel oil pressure of 125 pounds per square inch (lb/in^2). For atomizers specifically designated as light-off atomizers, the minimum capacity shall be not more than 200 pounds of oil per hour, or as specified (see 6.2). For type C burners, a stable flame pattern shall be maintained down to the specified minimum oil pressure. The low capacity at the minimum fuel oil pressure shall be not greater than the rated capacity at maximum oil pressure divided by the turndown (10 to 1).

3.2.11 Limiting firing distances. The least permissible distances for a service (multiple burner) installation shall be determined for each type and size of burner (see 4.6.1.7) and shall govern the determination of the position of burners in any boiler, depending on the number of burners per boiler and number of boilers per fireroom (see 6.2). These limiting firing distances shall be not less than those specified in table I.

TABLE I. Limiting firing distances.

Centerline of burner to	Burner capacities in pounds of oil per hour			
	Up to and including 2,100	From 2,101 to 3,300	From 3,301 to 6,000	Above 6,000
	Inches	Inches	Inches	Inches
Centerline of adjacent burners	22	25½	30	36½
Furnace floor	25	25	25	25
Nearest tube surface towards which gas is flowing	31	31	31	31
Water-cooled side wall	-	26	28	28
Side wall refractory	25	25	-	-

3.2.12 Air pressure drop across burner. The air pressure drop across the burner shall be as low as practical, consistent with the complete burning of the fuel within the available firing depth and the necessary mixing of the air and fuel oil. If a maximum allowable air pressure across the burner register at maximum burner capacity is required (see 6.2), the burner shall be designed to hold pressure drop at or less than this limit.

3.2.13 Fuel data. For construction and evaluation purposes, the following ultimate analysis, heating valves, and furnace heat releases shall be used:

Carbon	85.8 percent
Hydrogen	13.2 percent
Oxygen	0.1 percent
Sulphur	0.8 percent
Nitrogen	0.1 percent
Heating valve	19,500 British thermal units per pound (Btu/lb) of oil
Total furnace heat release at maximum capacity of burners per unit of furnace volume	700,000 British thermal units per hour per cubic foot ($Btu/hr/ft^3$) of furnace volume

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3.2.14 Threaded parts. Unless otherwise specified herein, or approved by the Technical Authority, all bolts, nuts, studs, and machine screws shall conform to FF-N-836, FF-S-85, FF-S-86, FF-S-92, FF-S-200, or MIL-DTL-1222, as applicable.

3.2.15 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Type A – straight pressure atomizing fuel oil burners. Type A burners are no longer in use except as a modification of type C atomizing burners (see 3.4.3). These utilized the fuel oil pressure mechanical atomizing principle exclusively, all oil supplied to these burners is atomized (no return flow). They shall operate at a fuel oil pressure range of 150 to 300 lb/in^2. However, emergency operation down to a fuel oil pressure of 125 lb/in^2 is permitted, and fires shall remain stable at this supply pressure.

3.4 Type C – steam and oil pressure atomizing burners. Type C burners shall employ steam as the atomizing medium for normal operation. Atomizers shall operate with steam conditions of 135 to 150 lb/in^2, 370 to 410 °F, and oil pressures of 350 to 35 lb/in^2 at the supply headers. They shall have a satisfactory turndown operating range of not less than 10 to 1 based on oil flow. The burner atomizer for type C burners shall be of the internal mix type and contain an outer barrel and a central tube. In order to accommodate the two lapped metal-to-metal seals on the sprayer plate (see 6.5 and Drawings 4168 and 221-7022537), the outer barrel and its seal at the atomizer head end shall be of a floating design. The atomizer shall also be capable of operating with 70 to 90 lb/in^2 air as the atomizing medium. This mode of operation shall provide an alternate means for lighting-off, and combustion shall be satisfactory with atomizing air up to a capacity of at least 5 percent of the rated boiler full power oil flow.

3.4.1 Oil/steam ratio. Atomizers shall atomize oil at the rate of not less than 7.7 pounds of oil for each pound of atomizing steam used when the burner is operated at firing rates in excess of 30 percent of the capacity of the atomizer.

3.4.2 Sprayer plates. Sprayer plates shall be designed for turndown ratios of 10 to 1. One plate size shall be provided to operate the boiler from 10 percent to 100 percent of the boiler full power rating based on oil flow (all burners firing). A second plate shall be provided that will fire the boiler from 100 percent to 120 percent of boiler full power rates. This is referred to as overload capacity. Sprayer plates capable of firing the boiler at overload rate shall be sized to prevent the oil flow from exceeding the 120 percent rate by more than 5 percent. Designs may exceed the specified turndown by reducing low end capacities, provided that the fires remain stable at minimum oil pressure. If a single burner cannot fire the boiler at rates less than 5 percent of the total boiler full power oil rate, then light-off sprayer plates shall be as specified (see 6.2), in order to permit light-off and raising of boiler pressure at the prescribed rate and time, without the need for securing fires.

3.4.3 Straight mechanical adaptors. Straight mechanical adapters should be provided that convert the atomizer to straight pressure atomizing (atomizer type A) (see 6.5).

3.5 Designation and marking.

3.5.1 Identification and designation plates. Identification plates and other designating marking, in accordance with MIL-DTL-15024 and MIL-P-15024/5, shall be provided. These plates shall be installed on and furnished as a part of the burners. They shall be mounted on the front of the burner so as to be located in a readily accessible position where they can be read at any time without danger to personnel.

3.5.2 Identification plate markings. Data marked on identification plates shall include the following:

a. Manufacturer’s name, identification, symbols, and serial number.

b. Contract number and date of manufacture.

c. Type and class of burner.

d. Blank space for Government inspector’s official stamping.

3.5.3 Designation plates. Each burner shall have a designation plate, bearing a number or letter, mounted on the burner front to designate the lighting-off sequence.

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3.5.4 Sprayer plate stamping. Sprayer plates shall be stamped with a size designation indicating the following:

a. Number of exit holes.

b. Cone angle between the exit holes.

c. Drill size of the exit holes.

d. Drill size of the steam holes.

e. Drill size of the oil holes.

3.6 Special tools. Special tools if needed shall not be required to install or service the burners. Special tools are defined as those tools not listed in the GSA Global Supply Catalog.

3.7 Workmanship. Workmanship shall be of uniform quality, free of molding fabrication defects, loose materials, damaged or improperly assembled burners, galling of mating parts, nicks or burrs of metal parts, or any post molding warpage.

3.8 Sprayer plate materials and hardness. Type A and C sprayer plates shall be manufactured from corrosion resisting steel in accordance with SAE-AMS-QQ-S-763, class 440C, condition A. An acceptable alternate material is T-1 tool steel in accordance with ASTM A600. Type A and C sprayer plates will be hardened to a range of Rockwell Hardness Scale C (HRC) 58 to 62 in accordance with the procedures of SAE-AMS-H-6875.

3.9 Shock. Naval burners shall withstand high-impact shock in accordance with MIL-DTL-901. Burners shall be fully operable without steam, fuel, or combustion gas leakage after shock exposure.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see table II and 4.3).

b. Conformance inspection (see table II and 4.4).

TABLE II. First article and conformance inspection.

Characteristics	Requirements	First article	Conformance
Construction, workmanship, materials	3.2 through 3.2.8, 3.2.14, 3.2.15, 3.7, 3.8	4.5, 4.6.1.4, 4.6.1.5, 4.6.3, 4.6.4, 4.6.5	4.5, 4.6.3, 4.6.4, 4.6.5
Capacities	3.2.8, 3.2.9, 3.2.10, 3.2.11, 3.2.12	4.6.1, 4.6.1.1, 4.6.1.2, 4.6.1.3, 4.6.1.4, 4.6.1.7	-
Atomization and combustion	3.4, 3.4.1, 3.4.2, 3.4.3	4.6.1.6	-
Marking	3.5, 3.5.1, 3.5.2, 3.5.3, 3.5.4	4.5	4.5
Shock	3.9	4.6.2	-
Repair parts and accessories	When so specified, 6.5	4.5	4.5

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in a laboratory with a burner test stand and an operating boiler. The environmental conditions shall be the same as those aboard ship except that ship motion is not required. Fuel analysis shall be in accordance with 3.2.13.

4.3 First article inspection. First article inspection shall be performed on each burner assembly when a first article sample is required (see 3.1). This inspection shall include the examination of 4.5 and all the tests of 4.3 as specified in table II.

4.4 Conformance inspection. Conformance inspection shall include the examination of 4.5 and the tests of 4.6.

4.5 Examination. Each burner shall be visually and dimensionally examined to determine conformance to the requirements of this specification. Failure of these criteria shall be considered basis for rejection of items.

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4.6 Tests. Each burner and all parts shall be tested to determine conformance to the requirements of this specification.

4.6.1 Capacities. The overload, low, and rated capacities (see 4.6.1.1, 4.6.1.2, and 4.6.1.3) in pounds of oil fired per burner per hour shall be demonstrated in a boiler or test facility. Air pressure drop across the burner over its operating range shall be measured to verify conformance to 3.2.12.

4.6.1.1 Overload capacity. Overload capacity shall be the amount of fuel oil that can be satisfactorily burned per hour per burner when the volume percent of carbon dioxide (CO_2) in the combustion gas is one unit less than the maximum as determined by a smooth performance curve based on test data. Loss of ignition, excessive carbon laying, improper flame conditions, or necessity for difficult adjustments shall be evidence of unsatisfactory firing.

4.6.1.2 Rated capacity. Rated capacity of the burners shall be the amount of fuel oil that can be satisfactorily burned per hour per burner in a multiple burner installation (the number of burners to be based on the typical installation shown in 803-5959293) at visible smoke condition of no more than Ringelmann scale 1 with a draft loss (pressure drop from fireroom or double front to the furnace) of 8 inches of water. The rated capacity is intended for comparative purposes only.

4.6.1.3 Low capacity. Low capacity (see 3.2.10) shall be demonstrated with one burner without difficult special adjustments. Operation at this capacity shall be at a visible smoke condition of no more than Ringelmann scale 1 and shall be carbon free or with such small carbon deposits (as built up in at least 4 hours of continuous operation) as not to affect burner performance or result in a coating on furnace refractories.

4.6.1.4 Operating range. The burners shall operate over the full operating range from low to overload capacity in smokeless condition (Ringelmann scale zero) and shall function under all operating conditions without depositing carbon to such extent as to affect the burner performance or result in a coating on furnace refractories.

4.6.1.5 Disassembly. After a specified period of boiler operation, an atomizer shall be selected, secured, and removed from the boiler. Removal shall be accomplished without binding. The atomizer shall be permitted to cool at ambient conditions for 15 minutes after which the cap nut and sprayer plate shall be removed with the normal removal tools. There shall be no binding of parts. Any tools required shall be identified to ensure compliance with 3.6.

4.6.1.6 Oxygen (O_2) and CO_2. During operational tests, the volume percentages of O_2 and CO_2 in the exit combustion gases shall be measured. At the maximum capacity, O_2 shall be 3.5 percent or less. At a fuel flow rate of 40 percent of boiler full power, the O_2 shall be 7 percent or less. Oxygen readings at other rates shall lay on a smooth curve. Equivalent CO_2 percentages for naval distillate fuel operation shall be 12.7 percent or greater at maximum capacity and 10 percent or greater at 40 percent of boiler full power.

4.6.1.7 Limiting firing distances. The limiting firing distances for each burner shall be determined at the overload capacity.

4.6.2 Shock. Burners shall be subjected to high-impact shock testing in accordance with grade A, class I, type A of MIL-DTL-901. Burners shall be fully operable, with no steam, fuel, or combustion gas leakage, after shock test completion.

4.6.3 Integrity/leakage tests. Each safety shutoff device and atomizer assembly shall be hydrostatically tested for integrity to 150 percent of the maximum system operating pressure and shall be tested for leakage at 100 percent of the maximum system operating pressure.

4.6.4 Safety shutoff device valve operation. Valves and spring check valves on each safety shutoff device shall be operated in the manner in which they will be used for normal and also emergency boiler operation to ensure compliance with the requirements of 3.2.2.

4.6.5 Hardness tests. Sprayer plates shall be tested for hardness in accordance with ASTM E18 to verify conformance to 3.8.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The burners covered by this specification are intended for use on propulsion boilers aboard naval ships. Current stock of burners may be used until depleted; future requisitions for burners should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and class required (see 1.2).

c. When first article inspection is required (see 3.1 and 6.4).

d. Dimensions for air registers (see 3.2.6).

e. Sizes and capacities of sprayer plates required and the quantity for each if other than as specified (see 3.2.10 and 6.5).

f. Number of burners per boiler and number of boilers per fireroom (see 3.2.11).

g. Maximum air pressure drop across burner register, if required (see 3.2.12).

h. Capacity range for light-off sprayer plates, if required (see 3.4.2).

i. Packaging requirements (see 5.1).

j. Requirements for technical manuals and other recommended documentation (see 6.3).

k. Requirements for accessories, repair parts, and tools (see 6.5).

6.3 Technical manuals and other recommended documentation. When specified (see 6.2), technical manuals and other documentation, to include the following, should be provided:

a. Test and material reports.

b. Drawings.

c. Provisioning technical documentation.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.6. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.5 Accessories, repair parts, and tools. When specified (see 6.2), accessories, repair parts, and tools required for burner maintenance should be furnished. Tools should be those available in the GSA Global Supply Catalog.

a. Suggested accessories:

(1) Atomizer assemblies, less sprayer plates: 250 percent of number of burners purchased.

(2) Sprayer plates: 300 percent of each size sprayer plate intended for use per burner for sizes and capacities procured (see 6.2).

(3) Adapters, steam to mechanical: 200 percent of number of type C burners purchased.

b. Suggested repair parts and special tools:

(1) Diffuser (also called impeller) plates: 25 percent of the number installed or ordered.

(2) Gaskets or seals of each size and type used for oil or steam tightness: 200 percent of the number installed or ordered.

(3) Gaskets of each size and type used for air tightness: 10 percent of the number installed or ordered.

(4) One set of wrenches required for sprayer plate assembly/disassembly in the atomizer for each boiler on order or as specified (see 6.2).

(5) One set of tools required for safety shutoff device seal replacement for each fireroom.

6.6 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.7 Subject term (key word) listing.

Air register

Atomizer

Safety shutoff device

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4530-2023-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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